

Application of Linear Algebra

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https://www.math.ucdavis.edu/~daddel/linear_algebra_appl/Applications/applications.html

Applications of Linear Algebra

- Application 1: Constructing Curves and surfaces passing through Specified points
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- Application 2: Least Square approximation
- Application 3: Traffic Flow
- Application 4: Electrical Circuits
- Application 5: Determinant
- Application 6: Genetics
- Application 7: Graph Theory
- Application 8: Cryptography
- Application 9: Markov Chain
- Application 10: Leontief Economic Model

Example 1

Given the point $A = (3,5)$ and $B = (-1, 2)$ find the equation $ax + by + c = 0$ that passes through the points A and B .

Solution to Problem 1

Substituting the coordinates of the given points A , B and (x,y) in the equation $ax + by + c = 0$, we obtain the following linear system:

$$\begin{cases} ax + by + c = 0 \\ 3a + 5b + c = 0 \\ -1a + 2b + c = 0 \end{cases}$$

With the matrix equation $AX=B$ where A is the coefficient matrix $\begin{bmatrix} x & y & 1 \\ 3 & 5 & 1 \\ -1 & 2 & 1 \end{bmatrix}$, $X = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ and $B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Note that $AX = 0$ has a nonzero solution if and only if $\det(A) = 0$.

Use cofactor expansion along the first line to find $\det(A)$.

$$\det(A) = x \begin{vmatrix} 5 & 1 \\ 2 & 1 \end{vmatrix} - y \begin{vmatrix} 3 & 1 \\ -1 & 1 \end{vmatrix} + 1 \begin{vmatrix} 3 & 5 \\ -1 & 2 \end{vmatrix} = 0$$

or

$$4x - 4y + 11 = 0. \text{ That is } a=4, b=-4 \text{ and } c=11.$$